MPX · PD-298 · DESIGN REVIEW

MPx Squeeze v2

A pocket microplastic filter for PET bottles and taps

What it does, what the virtual test bench showed, which of our 22 technical requirements it meets and misses, and the **design FMEA** that opens the next CED stage: embodiment design.

Ø52 × 50 mm
outside the bottle

202 g
core unit, all 316L

1 µm
steel disc + 20 µm prefilter

99.5 %
of 1–100 µm particles caught*

* simulated on the virtual test bench; physical tests pending

CONCEPTUAL DESIGN ✓ → EMBODIMENT DESIGN (NOW)



Concepts chosen. The next CED stage is embodiment design

Our course puts **embodiment design** after concept selection, and names **FMEA** as its method for finding errors (lecture 2, slide 36). Pahl & Beitz §10.4 says a design FMEA checks that the functions in the requirements list are fulfilled.

LEC 3–5 ✓
Task clarification
Requirements list (22), SNPS, objective tree

LEC 6–8 ✓
Conceptual design
Function structure, ideation, morphological chart, 10 concepts

LEC 9 ✓
Simulation
Analytical + virtual test bench (this deck)

LEC 10 ✓
Evaluation & selection
Concepts 04, 07, 05 chosen; squeeze → v2

NOW
Embodiment design
Layout, form, scale drawings, FMEA, optimisation

NEXT
Detail design
Tolerances, fits, final drawings, process FMEA

LATER
Production
Production instructions, prototype ×3

EMBODIMENT DESIGN STEPS (LECTURE 2, SLIDE 36) AND WHERE V2 STANDS

✓ Overall layout: general arrangement, spatial compatibility	v2 drawing set S01–S06, STEP models, Ø52 × 50 mm envelope
✓ Preliminary form design: shapes, materials, processes	316L CNC parts, sintered 316L disc and prefilter, silicone seals; BOM ₹14,900/kit
◑ Apply all major criteria: technical, economic, environmental, aesthetic, ergonomic	Technical by simulation (slides 5–12). Cost at volume and grip ergonomics still open
◑ Develop with scale drawings and critical reviews	Drawings done; this deck is the first critical review
▶ **Use methods for identification of errors (e.g. FMEA)**	Draft design FMEA in this deck (slides 13–16): 20 failure modes, 12 critical
○ Apply methods for optimisation	v2.1 changes from the FMEA, then re-simulate with bench-measured values
○ Outcome: embodiment (layout) ready for detail design	After bench tests 1–7 confirm the model

SO: CAN WE DO AN FMEA NOW?

Yes, and it is the right next step. The layout and forms exist, and the simulation tells us how each part behaves, so we can rate occurrence and severity with numbers rather than guesses.

- Design FMEA: every part and use step (squeeze, tap, backflush, inspect)
- Risk number RN = O × S × D, each rated 1–10; **RN > 125 is critical** (Pahl & Beitz)
- Ratings here are a first draft from the simulation; the team should review them together, as P&B recommend
- Process FMEA (CNC, sintering, welding) belongs to detail design

Sources: CED lec02 sl. 36–37, lec06 process slide; Pahl & Beitz, Engineering Design 3e, §10.4

What the product does

It screws onto a 28 mm PET bottle (or a tap) and filters microplastics as you squeeze. The removable steel disc is also the sample: lift it out and look at what it caught under a phone microscope.



1 Fit

Screw onto any 500 mL or 1 L PET bottle (28 mm PCO 1881 neck). A tap adapter lets tap pressure do the work: **2.2 L/min at 1.5 bar**.

2 Squeeze

Water goes through the built-in **20 µm prefilter**, then the removable **1 µm sintered steel disc**. At 60 N, **27 squeezes** empty 500 mL: **115 s** of squeezing, about **170 s** with the pauses to let go. On release, an umbrella valve lets air back in.

3 Inspect

Lift the disc into the holder. Nile red stain + blue LED + a 140× clip-on lens shows plastic particles **down to 2 µm**.

4 Clean

Backflush with tap water through the spout: the cake comes off and **97 %** of the clean flow returns.

KEY NUMBERS

Size	Ø52 × 50 mm outside the bottle, +12.5 mm inside the neck
Mass	202 g core unit; accessories 326 g
Materials	316L body, sintered 316L filters, FDA silicone seals. No plastic in the water path
Prefilter	20 µm laminate cup, 17.2 cm², built in
Disc	Ø40 cassette, 11.6 cm² active: A 1 µm (default), B 1.4 µm, C 2.5 µm
Flow	0.26 L/min squeezed at 60 N; 2.2 L/min on a 1.5 bar tap
Capture	99.5 % of 1–100 µm particles by count; 99 % at 1–2 µm (simulated)
Cost	Prototype kit ₹14,900; 3 kits ₹44,700, inside the ₹50,000 cap

COVERS ALL THREE PARTS OF OUR SNPS

Remove
the filter on the bottle

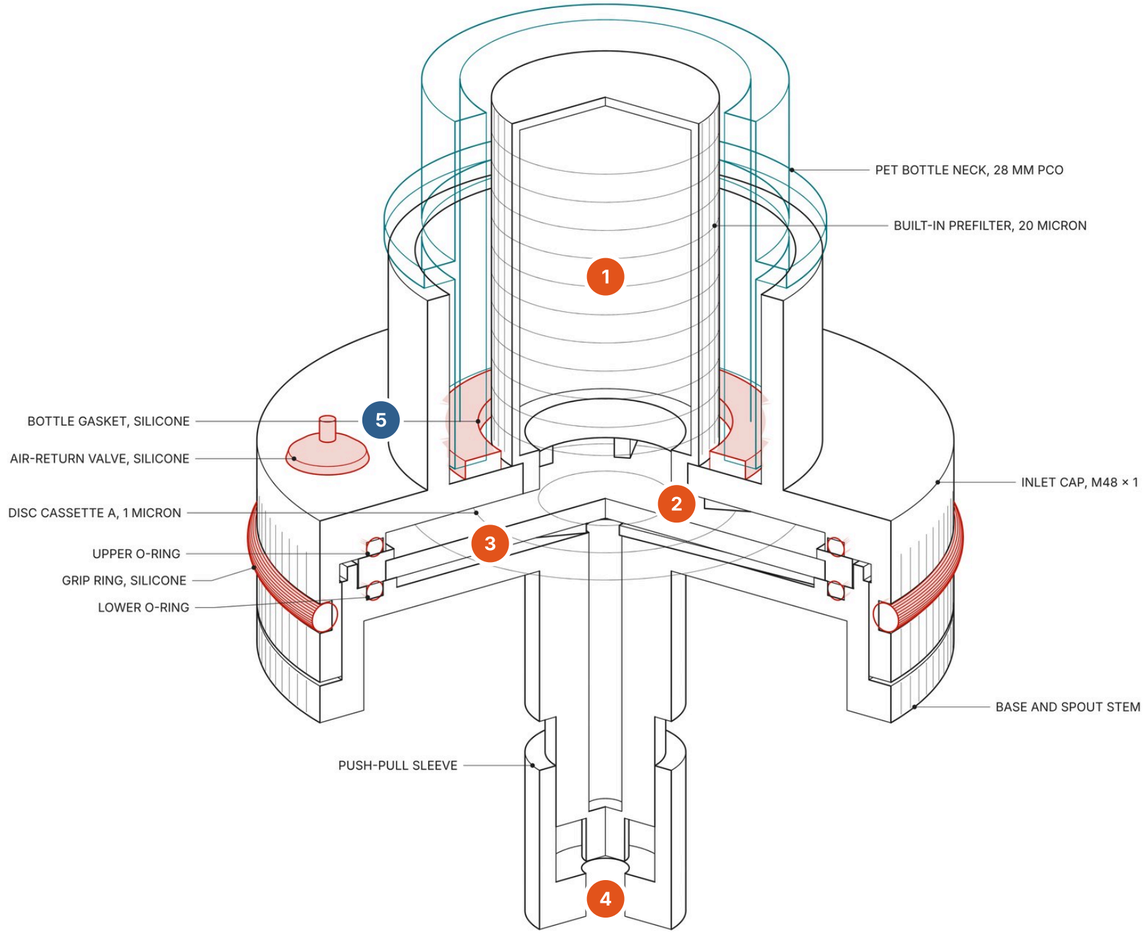
Detect
the disc is the slide

Contain
the catch stays on steel

Numbers from the v2 drawing set, BOM and the test-bench model (sq2 model.mjs)

How water moves through it

Two filter stages in series, sealed by two O-rings above and below the disc cassette. Those O-rings are the only barrier between dirty and clean water.



1 Into the prefilter

Bottle water passes outside-in through the 20 µm cup. Sand, fibres and plastic over 20 µm stay in the bottle side.

2 Over the disc

It spreads under 8 roof ribs across the full 38.5 mm disc face.

3 Through the 1 µm disc

The sintered 316L disc catches particles on its top face. The cake that forms there is what you inspect.

4 Out of the spout

Filtered water runs between 8 drain ribs to a Ø2 port in the stem and out through the push-pull sleeve.

5 Air back in

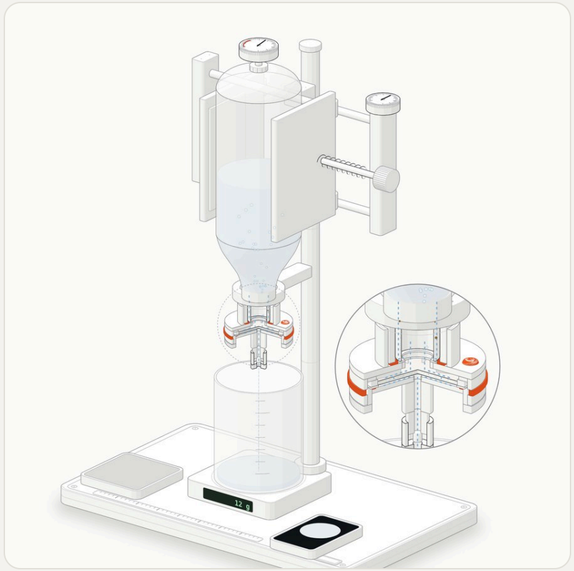
When you let go, the silicone umbrella valve opens at 0.03 bar and the bottle springs back. The wet disc cannot pass air: it needs 2.9 bar.

Not yet in the v2 drawings or BOM: the air-return valve and the silicone grip ring. Both are shown here and used in the simulation.

Cutaway: teardown drawing built from the v2 drawing-set profiles; threads omitted

The virtual test bench: four experiments

An interactive simulation of the bench we plan to build. A press squeezes the bottle with a set force and every number on screen comes from one physics model.



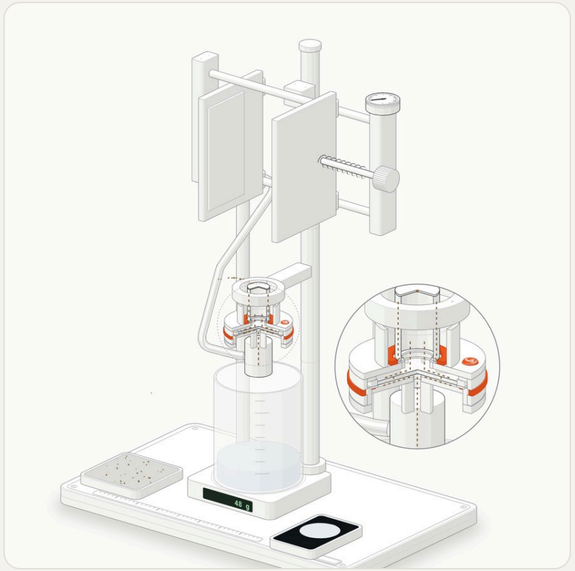
Squeeze, valve fitted

60 N on a 500 mL bottle gives **0.17 bar** and **0.26 L/min**. The bottle refills with air after each squeeze: 500 mL in **27 squeezes**, 168 s including release pauses.



Same squeeze, no valve

No air can get back in. Near **140 mL** the dented bottle holds **0.17 bar of vacuum**, equal to the squeeze, and flow stops.



Backflush

Tap water runs backwards: spout → disc → prefilter → out of the inlet. The cake lifts in about 9 s; **97 %** of clean flow returns.



Lift the cassette and look

The disc goes into the holder, cake up. With Nile red at 140×, particles from **2 µm** are visible (20 µm unstained).

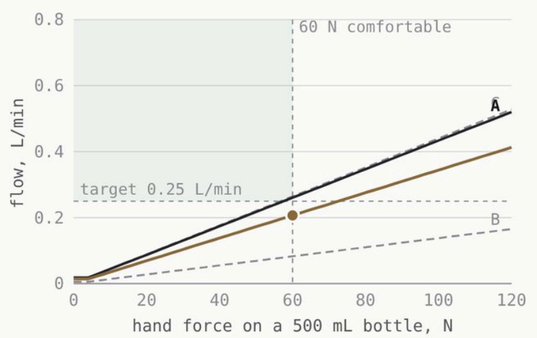
What the model includes: hand force → bottle pressure · Darcy flow through prefilter, cake and disc in series · water viscosity vs temperature · bottle vacuum without a valve · cake build-up and backflush · capture curve per particle size · Nile red visibility. Open it from Desktop › MPx › Squeeze_v2 › simulation.

What the bench reads out while it runs

Live charts, a requirements table that turns red or green as you change inputs, and the view through the phone microscope.

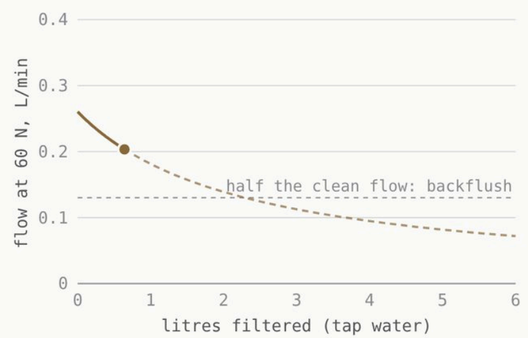
Length outside the bottle	50 mm	≤ 100 mm	pass
500 mL at 60 N, 25 °C	115 s	≤ 120 s	pass
Force for 0.25 L/min	58 N	≤ 60 N	pass
Bottle refills with air between squeezes	valve	every squeeze	pass
Air cannot leak back through the wet disc	2.9 bar	> 0.54 bar	pass
Capture 2–5 µm	100.0 %	≥ 99 %	pass
Capture 1–2 µm	99 %	≥ 90 %	pass
Capture below 1 µm	15 %	not claimed	–
Litres until flow halves, clean disc (tap)	2.2 L	≥ 20 L	fail
Flow recovered by backflush	98 %	≥ 80 %	pass
Tap flow at 1.5 bar	2.2 L/min	≥ 1.5 L/min	pass
Disc ΔP at that tap pressure	1.5 bar	≤ 3 bar	pass
Smallest particle seen on the disc	2 µm	≤ 5 µm	pass
Cap skirt safety factor at 3 bar	38×	≥ 4×	pass

Force needed for flow



cassette A clean — with today's cake — the other cassettes, at 25 °C. 0.25 L/min takes **73 N** now.

Flow as the disc loads



measured so far — ahead, if nothing changes — backflush. Flow halves in about **2.9 L** more of tap water, with the prefilter taking 40 % of the solids.

Phone microscope

Disc in holder (1–20 µm) | Backflush dish (prefilter catch + cake)

The cassette drops into the inspection holder with the cake facing up, so the disc itself is the slide. **633** particles of 1–100 µm are on it now. At this load a 0.64 mm field holds about **0.2** of them, so counting 100 means scanning about **572** fields. Nile red makes the plastic glow orange under a blue LED while the steel stays dark, so the lens limit applies.



140× · field 0.64 mm · Nile red, blue LED. Representative mix of the sample; **54** resolvable, 45 too small (under 2 µm).

HOW TO READ THEM

- **Force needed for flow:** cassettes A and C reach 0.25 L/min at 58 N; B (catalogue 0.2 grade) needs about 180 N.
- **Flow as the disc loads:** on tap water the flow halves after about 2.2 L. That is the one engineering check that fails.
- **Requirements table:** 12 pass, 1 fail (filter life), 1 not claimed (below 1 µm).
- **Microscope:** after 1 L there are only 0.1–0.2 particles per 0.64 mm field, so counting 100 particles means scanning about 560 fields.

The calculations behind the numbers

Eight hand-checkable equations. Each card shows the formula, our values, and the result the simulation uses.

1 Squeeze pressure

$$P = (F - F_{wall}) / A_{palm} + \rho \cdot g \cdot h$$

(60 − 4) N ÷ 34 cm² + 1000 × 9.81 × 0.126 m
= 0.165 + 0.012 bar

0.177 bar

Hand force spreads over the palm; the water column adds a little.

2 Flow, resistances in series

$$Q = \Delta P / [\mu (R_{disc}/A_{disc} + R_{pf}/A_{pf})]$$

Mott: K·t = 26 × 1.5 / 3 = 13 → R = 5.2×10^9 m^{-1}
μ(25 °C) = 0.89 mPa·s → 1.47 L/min per bar
0.177 bar × 1.47

0.26 L/min

115 s per 500 mL of squeezing; 168 s with release pauses.

3 Force for 15 L/h

$$F = (Q / q_1 - \rho g h) \cdot A_{palm} + F_{wall}$$

(0.25 / 1.471 − 0.012) bar × 34 cm² + 4 N
= 0.158 bar × 34 cm² + 4 N

58 N

Above the team's 50 N wish (5.1).

4 Why the valve is needed

$$P_{vac} = k \cdot \Delta V / V \;;\; P_{bubble} = 4\gamma / d$$

0.6 bar × 0.30 = 0.18 bar ≥ 0.177 bar squeeze
4 × 0.072 N/m ÷ 1 µm = 2.9 bar to push air back

stalls ~140 mL

Air can't return through the wet disc; a 0.03 bar umbrella valve fixes it.

5 When flow halves (cake)

$$R = R_m + \alpha \cdot m/A \;\rightarrow\; m_{½} = R_m \cdot A / \alpha$$

5.2×10^9 m^{-1} × 11.64 cm² ÷ 10^{13} m/kg = 0.60 mg
0.5 mg/L × (1 − 0.4) × 0.9 = 0.27 mg/L reaches the disc

2.2 L

Target 20 L: fails. Fix the prefilter.

6 Capture by size

$$E(d) = 1 / (1 + (d_{50} / d)^k)$$

Cassette A: d_{50} = 0.75 µm, k = 10
Particle count ~ d^{-2}, so the 1–2 µm band holds 51 % of all 1–100 µm particles

99.5 %

99.0 % at 1–2 µm; 15 % below 1 µm (not claimed).

7 Cap strength

$$\sigma_{hoop} = P \cdot r / t \;;\; SF = \sigma_y / \sigma$$

3 bar × 24 mm ÷ 1.6 mm = 4.5 MPa
170 MPa (316L yield) ÷ 4.5 MPa

SF 38

Wall is sized by machining and threads, not pressure.

8 Cold water

$$\mu(T) = 2.414 \times 10^{-5} \cdot 10^{\wedge}(247.8 / (T + 133.15))$$

μ(5 °C) = 1.52 mPa·s vs 0.89 mPa·s at 25 °C
flow ~ 1 / μ

−41 % flow

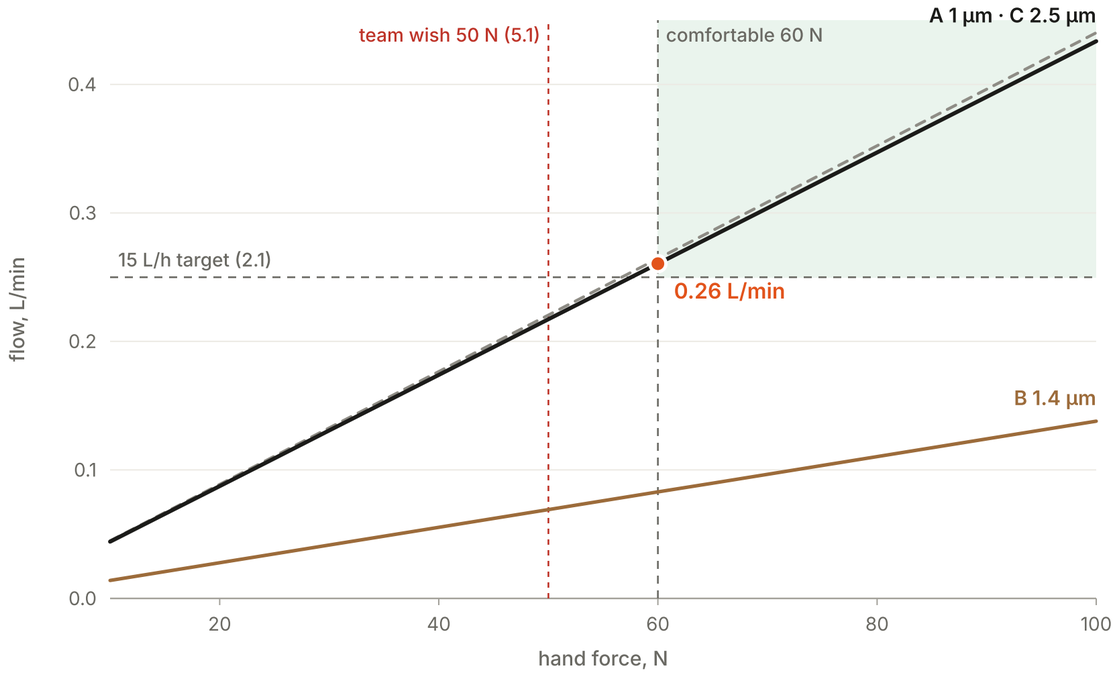
500 mL takes 194 s at 5 °C (8.2).

Inputs: v2 drawings (areas), Mott sintered-media K, assumed palm 34 cm², α = 10^{13} m/kg, 0.5 mg/L solids, prefilter takes 40 %

Result 1 · Flow: a 60 N squeeze meets 15 L/h only while squeezing

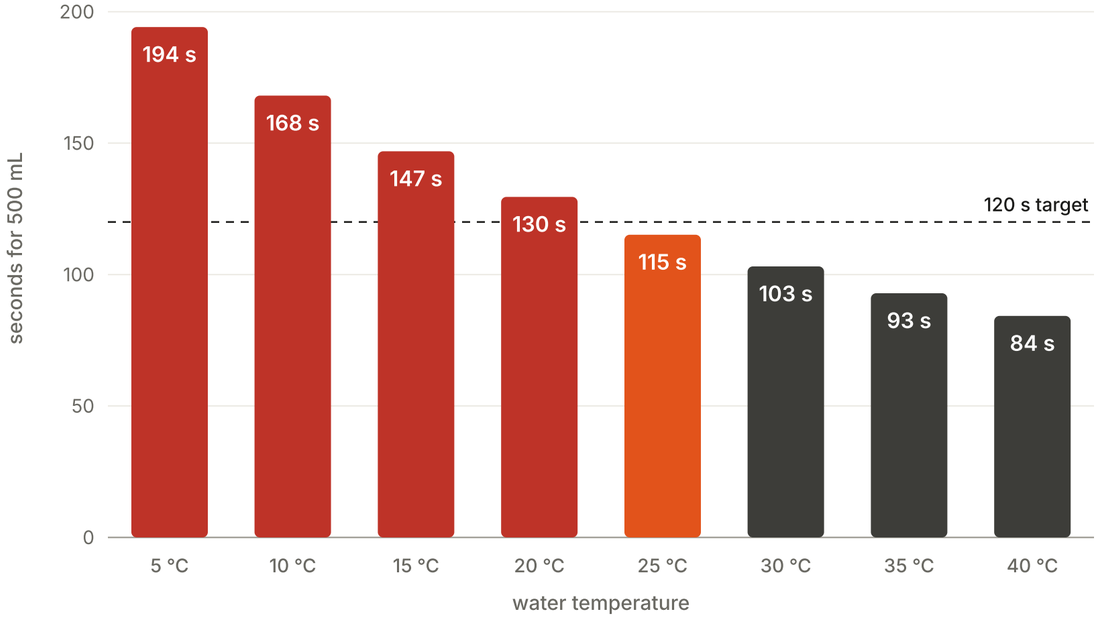
The asymmetric disc A is what makes this work: it flows like the coarse 2.5 µm disc C while filtering like a 1 µm membrane.

Flow vs hand force, 500 mL bottle, 25 °C



- **58 N** gives 0.25 L/min with A. **Passes** our 60 N comfort limit, **misses** the team's 50 N wish (5.1): 50 N gives 13 L/h.
- B, the catalogue 0.2-grade disc, needs about **180 N**. A is the highest-risk buy item, so the RFQ must ask for flux data.

Squeezing time for 500 mL vs water temperature, 60 N



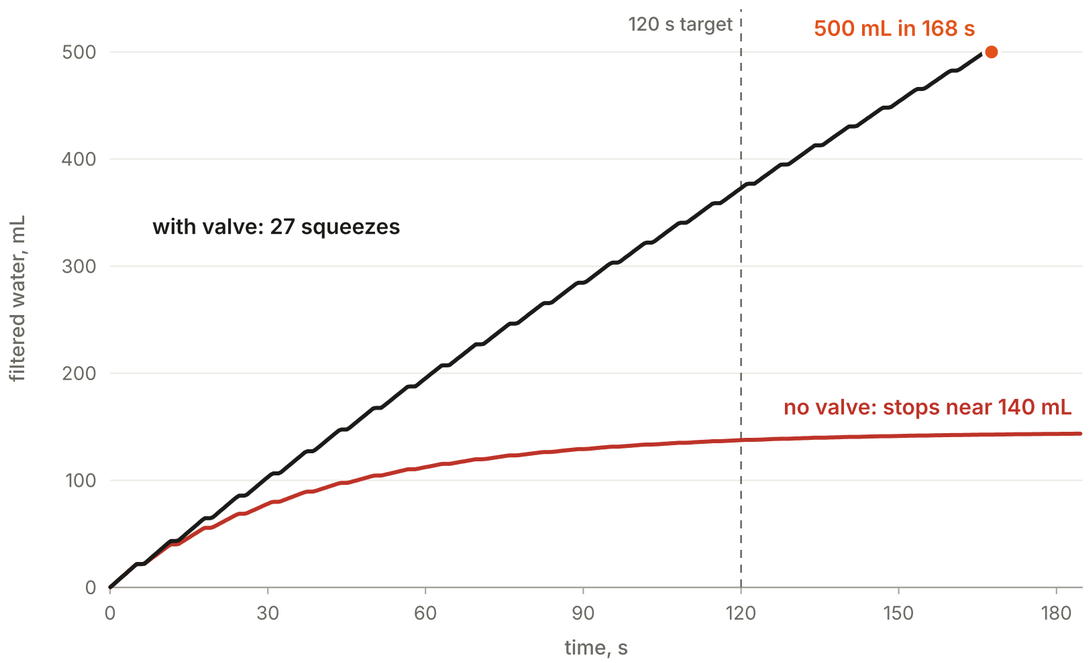
- Squeezing time only. At 25 °C, 500 mL takes **115 s**; below about **20 °C** it misses 120 s.
- Counting the 1.4 s release after each 5 s squeeze, 500 mL takes **168 s**: an average of **10.7 L/h**.
- At **5 °C** viscosity cuts flow by **41 %**. Usable over 5–45 °C (8.2), but slow.

Computed with model.mjs: flowLmin(handBar(F)), viscosity µ(T)

Result 2 · The air valve is essential, and the disc fouls fast

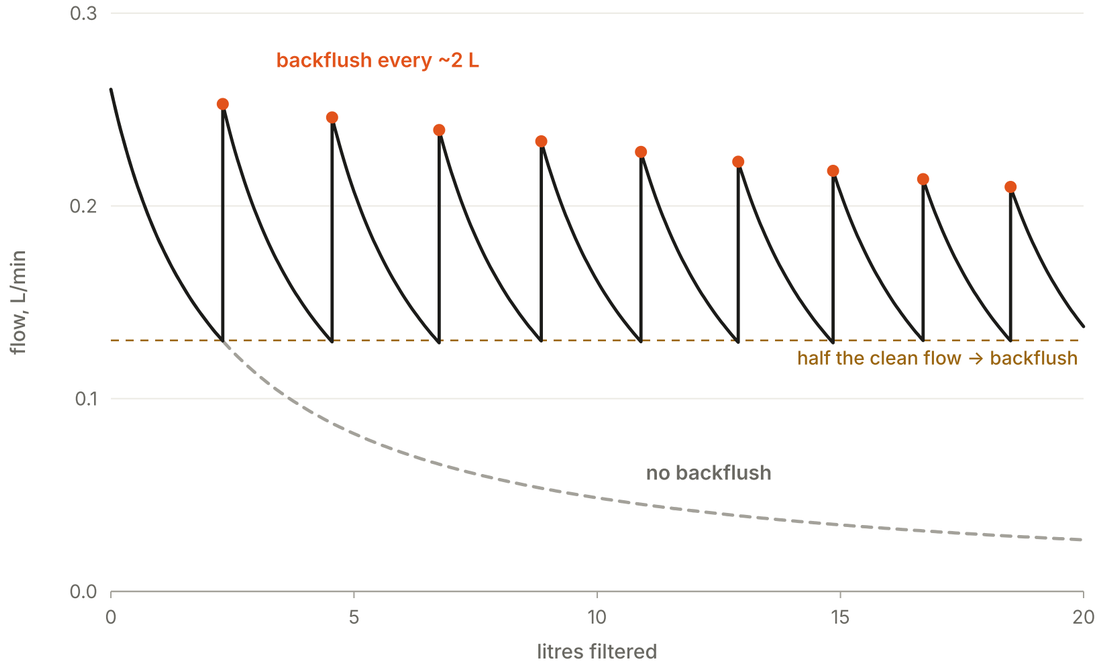
Without a way for air to return, a bottle stalls after one squeeze. With it, the limit becomes how quickly fine solids clog the 1 µm disc.

Water delivered vs time, 60 N squeezes, 500 mL bottle



- Without a valve, a dented bottle pulls **0.17 bar** of vacuum, as much as the squeeze. Flow stops near **140 mL**.
- Air can't come back through the wet 1 µm disc: that needs **2.9 bar** (bubble point). A **0.03 bar umbrella valve** solves it.

Flow at 60 N vs litres filtered, tap water (0.5 mg/L)



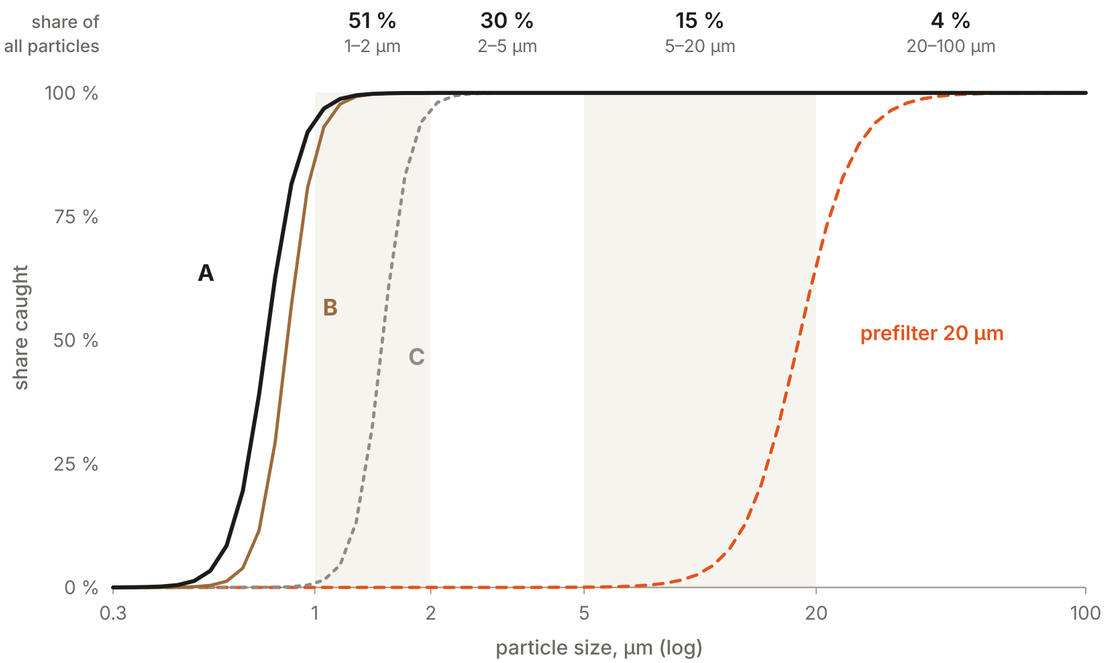
- Flow halves after **2.2 L** of tap water: **fails** our 20 L target and the team's 100 L filter rating (9.1).
- Backflush returns **97 %**, falling to **80 %** after 9 flushes (3 % stays each time). The disc is spent after **~75 L**.

Computed with model.mjs: stepSqueeze (valve on/off), cake filtration + stepBackflush

Result 3 · It catches what we claim; filter life needs a better prefilter

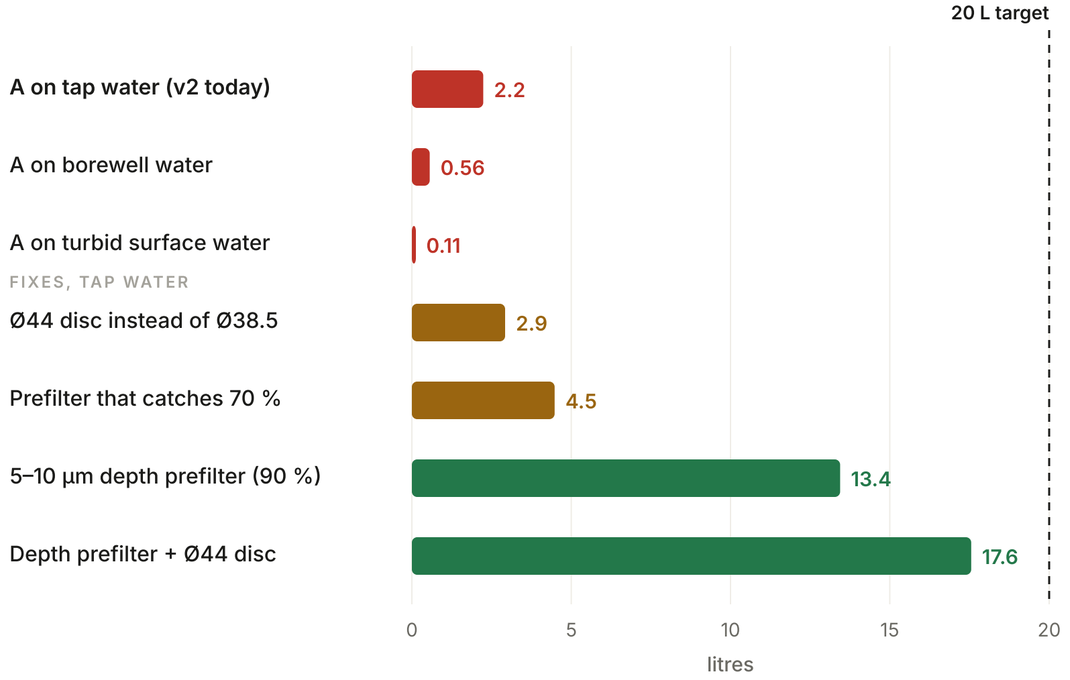
Capture is strong down to 1 µm. Filter life fails, and the simulation points at the fix: stop fine solids before they reach the disc.

Share of particles caught vs particle size



- Cassette A: **99.0 %** at 1–2 µm, **100 %** above 2 µm, **99.5 %** overall by count. **Passes** 6.1 including the 99 % stretch.
- Below 1 µm only **15 %**, so we don't claim nanoplastics. C (2.5 µm) lets **65 %** of 1–2 µm particles through.

Litres until flow halves (between backflushes)



- Today only 40 % of the solids stop at the 20 µm prefilter. A **5–10 µm depth prefilter** that stops 90 % gives **13 L** between backflushes.
- With a Ø44 disc as well: **18 L**. Borewell and turbid water need settling first.

Computed with model.mjs: bandCapture(), litresToHalf(); fixes scale disc area and prefilter share

Our 22 technical requirements: what v2 meets and what it doesn't

Ranked by our weighted objective tree. **Pass · sim**: the simulation shows it. **By design**: the choice should meet it, but nothing has tested it yet.

Pass · sim 1 | Met by design 3 | By design · untested 5 | Partial 4 | Fail 3 | At risk 1 | Not yet tested 4 | Not applicable 1

Rank	#	Requirement	v2 result	Checked by	Status
1	**6.1**	Removal ≥ 90 % by count, 1 µm–5 mm (stretch 99 %)	99.5 % overall; 99.0 % at 1–2 µm	Sim	Pass · sim
2	**3.2**	Filter medium sheds no secondary micro/nanoplastic	All-metal filters; silicone seals in the water path	Design	By design · untested
3	**6.2**	Detection false +/− ≤ 10 % vs reference microscopy	Sees ≥ 2 µm with Nile red; error rate not measured	Sim (partly)	Not yet tested
4	**10.2**	Captured microplastics stay contained	Backflush moves 97 % of the cake out; where it goes is up to the user	Sim	At risk
5	**4.1**	Result through ≥ 1 user feedback channel	Phone microscope view of the disc; manual count	Design	Partial
6	**11.1**	Retail price ₹1,500–6,000	Prototype ₹14,900/kit at qty 3; volume cost not estimated	BOM	Not yet tested
7	**3.1**	Water-contact parts food-contact safe	316L + FDA 21 CFR 177.2600 silicone	Spec	By design · untested
7	**3.4**	No leachables above IS 10500	Electropolished, passivated 316L (ASTM A967)	Spec	By design · untested
9	**2.1**	Flow rate ≥ 15 L/h	15.6 L/h while squeezing, 10.7 L/h with pauses; 132 L/h on a tap	Sim	Partial
9	**1.1**	Per-use capacity for 1–3 L/day	3 L ≈ 17 min of squeezing and pauses; backflush about daily	Sim	Partial
9	**10.1**	Disposal pathway for spent filters (W)	316L is recyclable scrap; process not defined	—	Not yet tested
9	**1.2**	Non-destructive sampling (W)	The disc itself is the slide	Design	Met by design
9	**5.3**	Display legible outdoors (W)	No own display; uses the phone	—	Not applicable
14	**9.1**	Filter rated ≥ 100 L or 3 months	Flow halves at 2.2 L; disc spent at ~75 L with backflush	Sim	Fail
14	**7.1**	Filter swap tool-free in ≤ 30 s	Unscrew by hand, lift cassette; not timed	Design	By design · untested
14	**4.2**	Alert near filter end-of-life	None; only slower flow	Design	Fail
14	**5.2**	Spout parts detachable and washable	Push-pull sleeve slides off the stem	Design	Met by design
14	**8.2**	Usable at 5–45 °C	Works; at 5 °C flow is 41 % lower (9.3 L/h)	Sim	Partial
19	**3.3**	Housing UV and impact resistant (W)	316L	Design	By design · untested
19	**3.5**	Survives 1 m drop (W)	Cap SF 38 at 3 bar; drop not tested	Calc	Not yet tested
21	**5.1**	Actuation force ≤ 50 N (W)	58 N needed for 15 L/h; 50 N gives 13 L/h	Sim	Fail
21	**8.1**	Noise ≤ 60 dB(A) (W)	No motor	Design	Met by design

Tier 1 (rank 1–4): removal passes, but **shedding, detection accuracy and containment** are unproven. The three fails each get a design fix in the FMEA (slides 14–16).

Requirements: MPx_Weighted_Objective_Tree.docx (22 items, D/W). W = wish

What the simulation proves, and what only a bench can

12 of 14 bench checks pass, but under assumptions we chose. Each assumption maps to a bench test that will measure it.

VALIDATED BY THE SIMULATION (BENCH READOUT)

Length outside the bottle	50 mm	≤ 100 mm	pass
500 mL at 60 N, 25 °C	115 s	≤ 120 s	pass
Force for 0.25 L/min	58 N	≤ 60 N	pass
Bottle refills with air between squeezes	valve	every squeeze	pass
Air cannot leak back through the wet disc	2.9 bar	> 0.54 bar	pass
Capture 2–5 µm	100.0 %	≥ 99 %	pass
Capture 1–2 µm	99 %	≥ 90 %	pass
Capture below 1 µm	15 %	not claimed	–
Litres until flow halves, clean disc (tap)	2.2 L	≥ 20 L	fail
Flow recovered by backflush	98 %	≥ 80 %	pass
Tap flow at 1.5 bar	2.2 L/min	≥ 1.5 L/min	pass
Disc ΔP at that tap pressure	1.5 bar	≤ 3 bar	pass
Smallest particle seen on the disc	2 µm	≤ 5 µm	pass
Cap skirt safety factor at 3 bar	38×	≥ 4×	pass

12 pass	Size, flow at 60 N, force, valve, bubble point, capture 1–5 µm, tap flow, disc ΔP, visibility, cap strength
1 fail	Filter life: 2.2 L until flow halves vs 20 L
not claimed	Below 1 µm: 15 %, not claimed
caveat	The 120 s check counts squeezing time only; with release pauses 500 mL takes 168 s (10.7 L/h)
not modelled	Not modelled at all: seal leaks, thread wear, shedding, leaching, cost, user behaviour

ASSUMPTIONS TO REPLACE WITH MEASUREMENTS

Assumption	Value used	Drives	Bench test
Palm contact area, bottle wall stiffness	34 cm², 4 N	Squeeze pressure, force	2 · pressure vs hand force, 5 users, 2 bottles
Asymmetric disc permeability	K·t = 13 (3× gain)	Flow, force (6.1, 2.1)	1 · flux vs ΔP for A, B, C
Bottle spring-back vacuum	0.6 bar per volume share	Need for the valve	2b · sealed bottle, vacuum gauge
Cake resistance α, prefilter share	10^{13} m/kg, 40 %	Filter life (9.1)	3 · fouling on tap + borewell water
Irreversible fouling per backflush	3 %	Disc service life	4 · 10 backflush cycles
Capture curve d_{so}, sharpness	A: 0.75 µm, k = 10	Removal (6.1)	5 · PS bead challenge 1, 2, 5 µm
Nile red visibility	2 µm at 140×	Detection (6.2)	6 · count vs reference microscope
Seals, threads, drop	not modelled	Leaks, galling (3.5)	7 · leak, 1 m drop, 200 thread cycles

CED LECTURE 9: THREE KINDS OF SIMULATION

Analytical (slide 7) **Virtual (slides 5–6)** **Physical: bench tests 1–7, next**

Bench tests 1–7 from the v2 specification; numbers in the readout are at default bench settings

Design FMEA: method

Pahl & Beitz §10.4: list how each part and use step can fail, what the user notices, and why. Then rate, rank and fix.

1 Risk analysis

Failures, consequences, causes; measures that avoid or detect them

2 Risk assessment

Rate occurrence O, severity S, and detection D (easy to detect = low D)

3 Risk number

RN = O × S × D, from 1 to 1000. **RN > 125 is critical**

4 Risk minimisation

Design change for each critical item, then re-rate for a new RN

Occurrence O

1–2	Unlikely: a design feature prevents it
3–4	Occasional
5–6	Likely over a year of use
7–8	Frequent: most users, often
9–10	Certain: the simulation shows it every time

Severity S

1–2	User barely notices
3–4	Slower or less convenient
5–6	Function degraded: hard squeezing, wrong count
7–8	Hygiene risk or contaminant released
9–10	Unfiltered water delivered as filtered

Detection D

1–2	Obvious at once (stall, drip)
3–4	User notices within a use
5–6	Found only by inspection or test
7–8	Hidden from the user
9–10	Cannot be detected

WORKED EXAMPLE: CASSETTE O-RINGS

Failure	Effect	Cause	Rating	RN	Action	New RN
Water bypasses the disc edge	Unfiltered water at the spout	O-ring twisted or missing on refit	O 4 × S 9 × D 8	288 · critical	Retained O-rings + spout blow-test	2 × 9 × 4 = 72

Scope: v2 core unit, the proposed air valve and grip ring, and four use modes (squeeze, tap, backflush, inspect). **20 failure modes, 12 critical.** Ratings use simulation numbers where they exist and judgement elsewhere: a draft for the team to review together, as P&B recommend.

Pahl & Beitz, Engineering Design 3e (2007), §10.4 and Fig. 10.11; CED lecture 2, slide 36

Design FMEA (1/2): the ten highest risks

Sorted by risk number. Red RN is critical (> 125). New RN is the re-rating once the action is in the design.

#	Part / function	Failure mode	Effect on user	Cause	Current control	O	S	D	RN	Recommended action	New RN	Req.
1	**Backflush output**	Cake washed into the sink	Caught microplastics go back to the environment	Backflush runs to the drain by habit; the dish is optional	Instructions only	8	6	8	384	Backflush into a lidded catch cup with a paper filter; dry and bin the paper	72	10.2
2	**Cassette O-rings S2/S3**	Water bypasses the disc edge	Unfiltered water at the spout; user can't tell	O-ring missing, twisted or pinched on refit; grit on the seat	Groove design, visual check	4	9	8	288	O-rings retained in the frame; blow-test at the spout (wet disc holds 2.9 bar)	72	6.1
3	**Clean side after backflush**	Clean side left full of tap water	First sips after a backflush are unfiltered	Backflush pushes tap water in through the spout	None	8	5	7	280	Squeeze 100 mL to waste after each backflush (mark on bottle), or backflush with filtered water	45	6.1
4	**Detection (Nile red)**	Wrong particle count	User trusts a false result	Nile red also stains organic matter and biofilm; < 2 µm unresolved	Visibility limit modelled only	7	6	6	252	H_2O_2 digestion step; calibrate with PS beads; compare with reference microscope	96	6.2
5	**Filter end-of-life**	No end-of-life warning	Keeps using a dead or damaged disc	No indicator; slow flow is the only cue	None	8	5	6	240	Litre and backflush log in the phone app; rule card: 500 mL > 4 min → backflush, 30 backflushes → replace	60	4.2
6	**Sintered disc, weld seam**	Crack or seam leak	Unfiltered water bypasses	Tap above 3 bar, water hammer, drop	3 bar rating	3	9	7	189	2 bar pressure limiter in the tap adapter; bubble-point test on delivery	72	6.1
7	**Cassette orientation**	Disc fitted upside down	Deep fouling, backflush fails, inspection face hidden	Frame fits both ways	Arrow mark	5	6	6	180	Stepped frame so it fits one way only	12	9.1
8	**Disc pores**	Irreversible fouling or biofilm	Flow never recovers; odour	3 % of each cake stays; wet storage	Backflush	6	6	5	180	Store dry; monthly boil or citric clean (316L takes 100 °C)	72	9.1
9	**Silicone seals, clean side**	Seal wear sheds particles	The filter adds particles downstream	Sliding S4 sleeve seal rubs every use	FDA silicone, post-cured	3	7	8	168	Static seals only on the clean side; blank-water run counts particles	70	3.2
10	**Disc fouling rate**	Flow halves after 2.2 L	Squeezing gets hard; user gives up	Fine solids pass the 20 µm prefilter (only 40 % stop)	Backflush returns 97 %	9	6	3	162	5–10 µm depth prefilter (13 L between flushes) + Ø44 disc	72	9.1

O occurrence · S severity · D detection, 1–10 each; RN = O × S × D. Draft ratings, for team review

Design FMEA (2/2): the remaining ten

Two more critical items (threads, spout hygiene); the rest are below 125 but still have cheap fixes.

#	Part / function	Failure mode	Effect on user	Cause	Current control	O	S	D	RN	Recommended action	New RN	Req.
11	**Cap M48 thread**	Galling, thread seizes	Can't open to swap or inspect the disc	316L on 316L, fine pitch, grit, no lubricant	None	5	7	4	140	M48 × 1.5; harder cap alloy or nitriding; food-grade silicone grease	56	7.1
12	**Spout and sleeve**	Clean side touched by dirty hands	Bacteria reach the mouth	Open spout, sleeve handled	Sleeve washable	5	7	4	140	Hinged dust cap on the spout	42	5.2
13	**316L surfaces**	Fe/Cr/Ni leaching or rust	Metals above IS 10500	Poor passivation, weld heat tint	ASTM A967 passivation	2	8	6	96	Electropolish; leach test to IS 10500	32	3.4
14	**Prefilter cup**	Clogs with silt	Very slow flow	Turbid surface water (10 mg/L)	Backflush clears it in ~1 s	6	5	3	90	Settle turbid water first; larger pleated cup	60	2.1
15	**Housing, disc frame**	Damaged by a 1 m drop	Dented frame leaks at the seals	Drop on a hard floor	Cap SF 38	3	6	5	90	Drop test ×10 (test 7); grip ring takes the impact	48	3.5
16	**Air-return valve**	Leaks while squeezing	Unfiltered drips near the spout	Damaged lip, wrong crack pressure	Visible drip	4	6	3	72	Valve away from the spout drip path; test 0.03 bar crack	36	6.1
17	**Bottle neck seal S1**	Leak at the bottle neck	Unfiltered water on hands and spout	Non-1881 necks on some Indian bottles	Gasket	6	6	2	72	Test 10 local bottle brands; thicker gasket	36	1.1
18	**Hand force**	User can't reach 60 N	Flow below 15 L/h	Weak grip: children, elderly	1 L bottle, tap mode	6	4	2	48	Grip ring; promote tap mode	32	5.1
19	**Air-return valve**	Sticks shut	Flow stalls after ~140 mL	Dried silicone, debris	Stall is obvious	4	4	2	32	Shield the valve with a rib; spare valve in kit	16	2.1
20	**Cold water**	Flow 41 % lower at 5 °C	500 mL takes 194 s	Viscosity	None	4	3	2	24	Note in the user guide	24	8.2

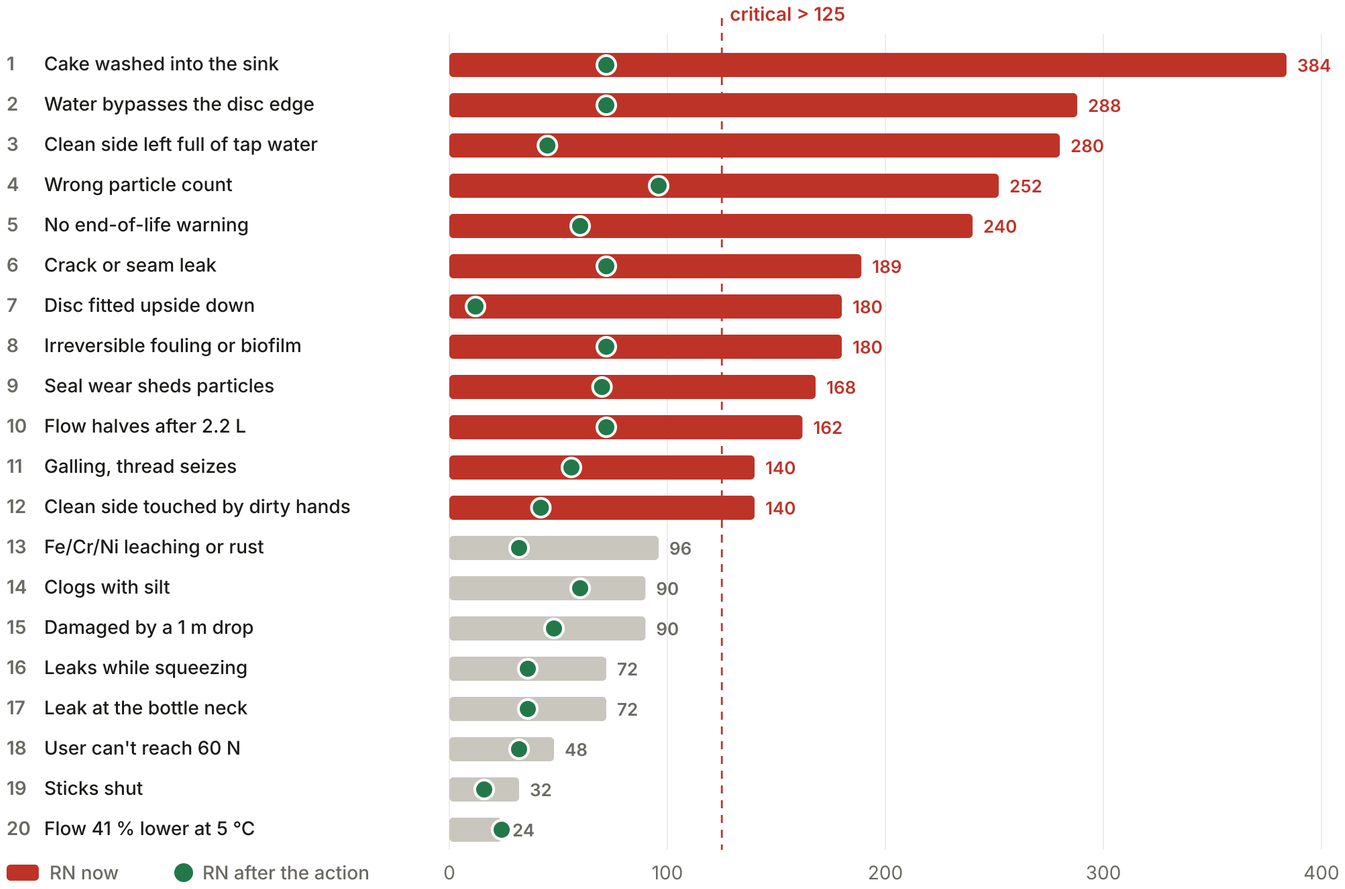
Not covered here: the process FMEA for CNC turning, sintering, laser welding and passivation. Pahl & Beitz treat it separately; it belongs to detail design, once suppliers and processes are fixed.

O occurrence · S severity · D detection, 1–10 each; RN = O × S × D. Draft ratings, for team review

FMEA result: twelve critical risks, all fixable in v2.1

The biggest risks are not the filter itself but what happens around it: **backflush, reassembly and detection**.

Risk number per failure mode, before → after



CHANGES FOR V2.1, IN PRIORITY ORDER

1. **Backflush catch cup**
   Lidded cup with a paper filter so the cake is kept, dried and binned (10.2)
2. **Flush-to-waste step**
   100 mL squeezed to waste after every backflush
3. **Keyed, sealed cassette**
   Stepped frame fits one way; O-rings retained; spout blow-test
4. **Depth prefilter + Ø44 disc**
   5–10 µm depth stage: 13–18 L between backflushes (9.1)
5. **Detection protocol**
   H_2O_2 digestion, bead calibration, reference count (6.2)
6. **Life log in the app**
   Litres and backflush count → end-of-life alert (4.2)
7. **Anti-galling thread**
   M48 × 1.5, harder cap, food-grade grease
8. **Tap pressure limiter, dust cap, static clean-side seals**

Two of the top three risks come from backflush. It cleans the disc, but it is also where the catch can escape (10.2) and unfiltered tap water reaches the clean side. Fixing backflush first takes those two from 664 risk points to 117.

All 20 failure modes re-rated below 100 after the actions; ratings to be confirmed by bench tests 1–7

Next steps: finish embodiment design, then detail design

The model told us where to look. Bench tests turn its assumptions into measurements, and the FMEA turns its risks into design changes.

WEEK 1

Review the FMEA as a team

- Agree O/S/D ratings
- Confirm the 12 critical items
- Assign owners

WEEK 1

Order parts

- RFQ for the asymmetric disc with flux and bubble-point data
- B and C discs as fallback
- 3 prototype kits (₹44,700)

WEEK 1–2

v2.1 layout

- Catch cup + flush step
- Keyed cassette, Ø44 disc
- Depth prefilter, valve, grip ring into drawings and BOM

WEEK 3–6

Bench tests 1–7

- Physical simulation (lecture 9) of every assumption on slide 12

WEEK 6

Close the loop

- Measured values into the model
- Re-run the requirements table
- Re-rate the FMEA

THEN

Detail design

- Tolerances and fits
- Final drawings
- Process FMEA: CNC, sintering, welding, passivation

BENCH TESTS: WHAT EACH ONE MEASURES, AND THE REQUIREMENT IT SETTLES

1 Flux vs ΔP, cassettes A, B, C
Measures K·t of each disc
settles 2.1, 5.1

2 Bottle pressure vs hand force, 5 users
Measures palm area, wall stiffness, vacuum
settles 2.1, 5.1

3 Fouling on tap and borewell water
Measures cake resistance α, prefilter share
settles 9.1, 1.1

4 Backflush recovery, 10 cycles
Measures irreversible share per flush
settles 9.1, 10.2

5 Polystyrene bead challenge, 1/2/5 µm
Measures real capture curve
settles 6.1

6 Nile red count vs reference microscope
Measures false positive / negative rate
settles 6.2, 4.1

7 Leak, 1 m drop, 200 thread cycles
Measures seal and thread life
settles 3.5, 7.1

+ Blank-water shedding run; leach test
Measures particles from seals; metals vs IS 10500
settles 3.2, 3.4

Bench tests 1–7 from the v2 specification; detail design follows the embodiment outcome (lecture 2, slide 37)